

Absolutely. Below is the **master content** you can use to prepare both your **long PDF report** and your **20–22 slide PPT**.

I have aligned it with the **SecureCorp instructor handbook**, the instructor summary you gave me, and the lab work we actually completed. The handbook specifically requires the Build → Discover → Attack → Log → Detect → Investigate → Assess → Remediate → Report methodology, evidence, attack logs, detection notes, incident timeline, recommendations, executive summary and presentation. Home_lab_setup

I am also keeping the report **technically honest**: where Wazuh target-side monitoring was not established, we will say **Not Visible / Not Verified**, rather than inventing alerts. Your earlier generated report also followed this evidence-based approach.
SecureCorp_COMPLETE_FINAL_PROJ

PART A — COMPLETE PDF CONTENT

COVER PAGE

SECURECORP

CYBERSECURITY SECURITY ASSESSMENT

Junior Cybersecurity Consultant – Four-Week Capstone

Final Security Assessment Report

Prepared by: Navya.k

Role:

Junior Cybersecurity Consultant

Assessment Type:

Authorized Cybersecurity Training Laboratory

Environment:

Ubuntu WSL • Metasploitable2 • DVWA • Wazuh Cloud

Assessment Methodology:

Build → Discover → Attack → Log → Detect → Investigate → Assess → Remediate → Report

Submission Date: 09-10-2026

PAGE 2 — CONFIDENTIALITY / DOCUMENT CONTROL

Document Control

Field	Details
Organization	SecureCorp Inc.
Project	Cybersecurity Assessment Capstone
Consultant	Navya.k
Role	Junior Cybersecurity Consultant
Assessment Type	Controlled Security Laboratory
Primary Target	Metasploitable2 + DVWA
Analyst Platform	Ubuntu WSL
Monitoring	Wazuh Cloud
Report Version	1.0
Status	Final Academic Submission

Confidentiality Statement

This report has been prepared exclusively for the SecureCorp cybersecurity capstone assessment. The activities described were performed within an authorized cybersecurity training laboratory. No public, third-party or unrelated systems were intentionally targeted.

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1. EXECUTIVE SUMMARY

SecureCorp's cybersecurity environment was assessed as part of a four-week Junior Cybersecurity Consultant capstone engagement. The objective was to evaluate the security posture of an intentionally vulnerable laboratory environment from both offensive and defensive perspectives.

The assessment followed the security operations lifecycle:

Build → Discover → Attack → Log → Detect → Investigate → Assess → Remediate → Report.

The laboratory consisted of an Ubuntu WSL environment used as the attacker and analyst workstation, a Metasploitable2 vulnerable Linux server hosting multiple intentionally insecure services, the Damn Vulnerable Web Application (DVWA), and Wazuh Cloud for security monitoring.

Network discovery identified a broad attack surface. The recorded service discovery identified FTP, SSH, Telnet, SMTP, DNS, HTTP, Samba, NFS, MySQL, PostgreSQL, VNC and X11-related services. Legacy versions were observed across several services. The exposed MySQL service on port 3306 was particularly notable because the project baseline expected the database to remain inaccessible from the attacker network.

DVWA application testing was performed against the controlled target. The assessment covered the required application-security categories: brute-force authentication, SQL injection, cross-site scripting and command injection. The controlled FTP credential-testing exercise also used a self-created low-privilege account.

Wazuh Cloud was successfully configured and the **SecureCorp-WSL** agent was enrolled. The agent was running and reported a connected state. However, the Wazuh agent was deployed on the Ubuntu WSL analyst environment rather than directly on Metasploitable2. Therefore, target-side automatic detection of all Week 2 attacks cannot be claimed unless corresponding target-side telemetry and alerts are available.

The assessment demonstrates that the vulnerable target has a significantly larger attack surface than a hardened production server should normally expose. The highest-priority recommendations are to reduce exposed services, remove obsolete protocols, restrict database access, strengthen authentication, apply secure development controls, improve logging and monitoring, and establish complete target-side Wazuh visibility.

2. PROJECT BACKGROUND

SecureCorp is a fictional organization requiring a cybersecurity assessment to understand its exposure to network and application-based attacks.

The consultant's role is to simulate the activities of both:

Red Team

The red-team perspective asks:

"What could an attacker discover and exploit?"

Activities include:

- Host discovery
- Port scanning
- Service enumeration
- Web application testing
- Authentication testing
- Injection testing
- Credential testing

Blue Team

The blue-team perspective asks:

"Would defenders see the activity, and how would they investigate it?"

Activities include:

- Wazuh monitoring
- Log analysis
- Detection review
- Alert investigation
- Timeline construction
- Incident classification
- Remediation recommendations

The handbook describes this transition from attacker perspective to defender perspective as a central goal of the capstone. Home_lab_setup

3. PROJECT OBJECTIVES

The major objectives of the assessment were:

1. Build an isolated cybersecurity laboratory.
2. Configure the vulnerable Metasploitable2 target.
3. Configure DVWA for web application testing.
4. Configure Ubuntu WSL as the attacker/analyst system.
5. Identify the target IP address.
6. Discover exposed ports and services.
7. Build an asset inventory.
8. Perform CIA analysis.
9. Perform risk assessment.
10. Test required DVWA vulnerabilities.
11. Perform controlled FTP authentication testing.
12. Attempt controlled SSH authentication testing.
13. Deploy Wazuh monitoring.
14. Verify Wazuh agent health.

15. Investigate available security telemetry.
 16. Distinguish events, alerts and incidents.
 17. Build an incident timeline.
 18. Recommend practical security controls.
 19. Prepare an executive summary.
 20. Present the assessment professionally.
-

4. SCOPE AND RULES OF ENGAGEMENT

Authorized Scope

The assessment was limited to:

- Ubuntu WSL attacker/analyst environment
- Metasploitable2 target
- DVWA application
- Wazuh Cloud monitoring environment
- Associated laboratory network

Testing Restrictions

Only the authorized training target was assessed.

No:

- public websites
- third-party systems
- unrelated computers
- shared infrastructure
- other students' systems

were intentionally targeted.

Credential Testing

Password testing was restricted to the self-created low-privilege laboratory account:

Username: `securetest`

Password used for controlled validation: `Secure123`

A small password list was used to avoid uncontrolled credential attacks.

Evidence Requirement

Every significant finding should be supported by:

- Screenshot
- Terminal output
- Browser evidence
- Wazuh evidence
- Log evidence
- Timestamp
- Target information

The handbook specifically emphasizes complete evidence collection and timestamps.
Home_lab_setup

5. SECURITY ASSESSMENT METHODOLOGY

Security Operations Loop

1. Build

Create the laboratory environment.

2. Discover

Identify hosts, ports and services.

3. Attack

Perform controlled security tests.

4. Log

Capture timestamps and evidence.

5. Detect

Determine whether Wazuh detects the activity.

6. Investigate

Analyze logs and alerts.

7. Assess

Determine business and technical risk.

8. Remediate

Recommend security controls.

9. Report

Document and communicate findings.

This methodology is directly aligned with the handbook's project mission and operations loop. Home_lab_setup

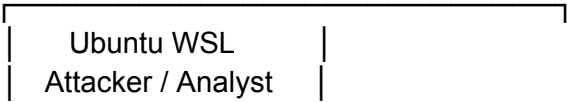
6. LAB ARCHITECTURE

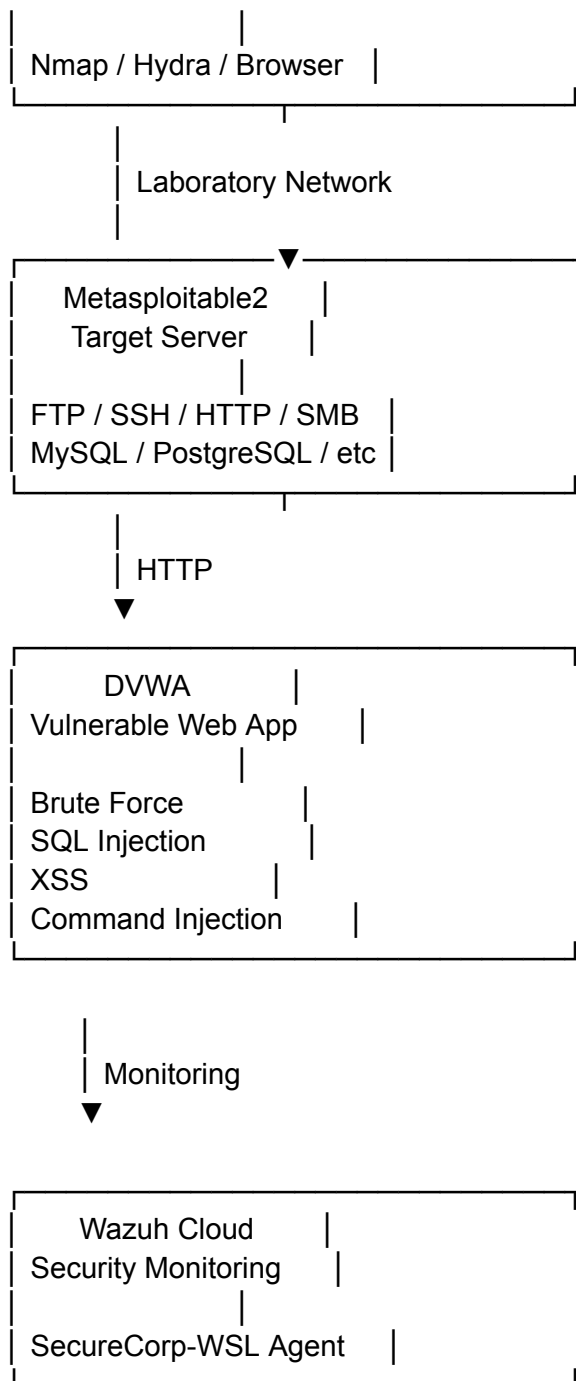
Components

Component	Role
Ubuntu WSL	Attacker / Analyst
Metasploitable2	Vulnerable Target
DVWA	Web Application Target
Wazuh Cloud	Security Monitoring
Wazuh Agent	Endpoint Telemetry
VirtualBox	Virtualization
Laboratory Network	Connectivity

Architecture Diagram

Put this diagram in the PDF:



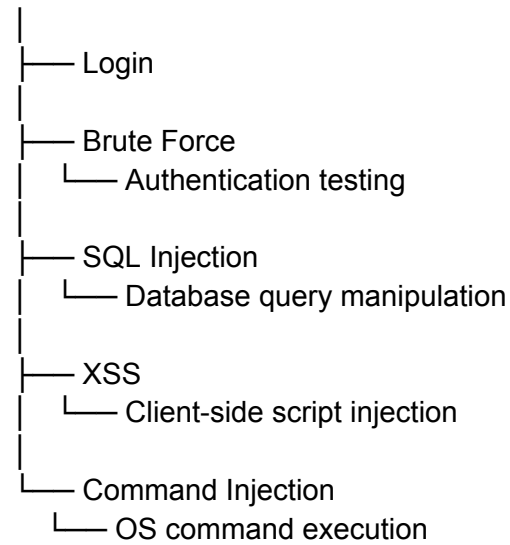


Important: In your actual report, clearly indicate that the verified Wazuh agent was on WSL rather than Metasploitable2.

7. APPLICATION MAP

DVWA Application

DVWA



Application Security Level

Low

This setting was intentionally selected because the capstone requires controlled demonstration of vulnerable application behavior.

8. ASSET INVENTORY

Asset	Type	Function	Importance
Metasploitable2	VM/OS	Vulnerable target	Critical
DVWA	Web application	Application testing	Critical
Ubuntu WSL	VM/OS	Attacker/analyst	High
Wazuh Cloud	Security platform	Monitoring	High
Wazuh Agent	Security software	Telemetry	High
Laboratory Network	Connectivity	Communication	High
Security Evidence	Data	Assessment evidence	High
Lab Credentials	Identity	Controlled authentication	High

9. CIA ANALYSIS

CIA means:

- Confidentiality
- Integrity
- Availability

Asset	Confidentiality	Integrity	Availability	Priority
Metasploitable2	High	High	High	Critical
DVWA	High	High	Medium	High
Ubuntu WSL	Medium	High	Medium	Medium
Wazuh Cloud	High	High	High	Critical
Security Logs	High	High	High	Critical
Lab Credentials	High	High	Medium	High
Network	Medium	High	High	High

Explanation

Confidentiality is important because unauthorized access could expose information.

Integrity is important because attackers modifying systems or databases could corrupt information.

Availability is important because disruption of critical systems could prevent legitimate operations.

10. RISK ASSESSMENT

Asset	Threat	Vulnerability	Impact	Risk	Recommendation
DVWA	Web attacker	SQL injection	Data disclosure	Critical	Parameterized queries
DVWA	Web attacker	XSS	User/session compromise	High	Output encoding
DVWA	Web attacker	Command injection	OS command execution	Critical	Input validation
FTP	Credential attacker	Weak authentication	Unauthorized access	High	Disable/restrict FTP

SSH	Credential attacker	Legacy configuration	Unauthorized access	High	Modern SSH
Telnet	Remote attacker	Clear-text protocol	Credential exposure	Critical	Disable Telnet
MySQL	Remote attacker	Exposed port	Database compromise	High	Restrict port 3306
VNC	Remote attacker	Exposed remote service	Remote access	High	Restrict/disable
X11	Remote attacker	Legacy exposure	Host compromise	High	Disable/restrict
Network	Remote attacker	Broad service exposure	Expanded attack surface	High	Firewall/segmentation
Wazuh	Monitoring gap	Target not monitored	Delayed detection	High	Install target agent

11. WEEK 1 — ENVIRONMENT SETUP

Objective

The purpose of Week 1 was to build and understand the assessment environment before performing attacks.

Activities

- VirtualBox installation
 - Metasploitable2 setup
 - Network configuration
 - Ubuntu WSL preparation
 - DVWA access
 - Target IP identification
 - Nmap discovery
 - Asset inventory
 - CIA analysis
 - Risk assessment
 - Wazuh deployment
-

12. TARGET IDENTIFICATION

The active target used during the later laboratory configuration was:

Target IP: 192.168.137.72

Metasploitable2 login:

Username: msfadmin

Password: msfadmin

The target was kept within the controlled laboratory environment.

Screenshot

Insert:

Real Screenshot – Metasploitable2 / target IP

Caption:

Figure 1: Metasploitable2 target environment and network configuration.

13. WAZUH DEPLOYMENT

A Wazuh Cloud environment named:

SecureCorp-Capstone

was created.

The Wazuh agent installed on Ubuntu WSL was:

SecureCorp-WSL

Version:

Wazuh Agent 4.14.8

The service was verified as running.

The local state also reported:

status='connected'

The Wazuh dashboard showed:

Active: 1
Disconnected: 0
Pending: 0
Never connected: 0

Screenshot

Insert real Wazuh screenshots here:

- Wazuh Cloud environment
- Agent deployment
- Agent installation
- Agent Active dashboard
- Agent details
- Local connected status

These Wazuh setup facts are also documented in the earlier report artifacts.
SecureCorp_COMPLETE_FINAL_PROJE...

14. WEEK 2 — RED TEAM ASSESSMENT

The purpose of Week 2 was to simulate controlled attacker activity against the authorized laboratory target.

Required activities:

1. Nmap discovery
 2. DVWA application mapping
 3. DVWA brute force
 4. SQL injection
 5. XSS
 6. Command injection
 7. FTP brute force
 8. SSH brute force
-

15. T2.1 — NMAP DISCOVERY

Objective

Identify exposed network services and understand the target attack surface.

Command

```
nmap -sV 192.168.137.72
```

The recorded service-version assessment identified a broad legacy service surface.

Major Results

Port	Service	Version
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH 4.7p1
23	Telnet	Linux telnetd
25	SMTP	Postfix
53	DNS	BIND 9.4.2
80	HTTP	Apache 2.2.8
139	SMB	Samba
445	SMB	Samba
2049	NFS	NFS
2121	FTP	ProFTPD 1.3.1
3306	MySQL	5.0.51a
5432	PostgreSQL	8.3
5900	VNC	VNC
6000	X11	X11

Security Assessment

The target exposes considerably more services than would normally be expected on a hardened production server.

The most significant exposure includes:

- Telnet
- FTP
- legacy SSH
- database services
- VNC

- X11
- SMB
- NFS

The exposed MySQL port is particularly important because the project baseline expected the database to remain restricted.

Evidence

Insert:

Real Nmap screenshot

Caption:

Figure 2: Nmap service-version discovery against the authorized SecureCorp target.

16. T2.2 — DVWA APPLICATION MAPPING

DVWA was accessed through the target HTTP service.

The application was configured at:

Security Level: Low

Required testing areas:

- Brute Force
- SQL Injection
- XSS
- Command Injection

Security Significance

DVWA provides a controlled environment for demonstrating common web vulnerabilities without targeting a real production application.

Evidence

Insert:

Real DVWA screenshot

17. T2.3 — DVWA BRUTE FORCE

Objective

Demonstrate the risk of weak authentication and insufficient login protections.

Test

The DVWA login functionality was tested using controlled credentials.

Examples tested included:

admin / 123456
admin / admin
admin / password

A successful authentication was obtained using the valid lab credential:

admin / password

The application displayed:

Welcome to the password protected area admin

Security Impact

A brute-force attack becomes more effective when an application lacks:

- Account lockout
- Rate limiting
- MFA
- CAPTCHA
- Strong password requirements
- Login monitoring

Risk

High

Recommendation

Implement:

- MFA
- rate limiting

- account lockout
- password complexity
- breached-password detection
- authentication logging

Evidence

Insert the real:

- failed login screenshot
- successful login screenshot

Caption:

Figure 3: Controlled DVWA authentication testing demonstrating weak credential protection.

18. T2.4 — SQL INJECTION

Objective

Test whether user-controlled input can alter the SQL query executed by the application.

Payload

' OR '1'='1' #

Observed Result

The application returned multiple records rather than a single expected record.

Observed entries included:

- admin
- Gordon Brown
- Hack Me
- Pablo Picasso
- Bob Smith

Security Impact

SQL injection may allow attackers to:

- bypass application logic
- retrieve unauthorized records
- modify database content

- extract sensitive information
- potentially compromise the underlying database

Risk

Critical

Root Cause

The vulnerability is consistent with unsafe construction of SQL queries using untrusted user input.

Recommendation

Use:

- Prepared statements
- Parameterized queries
- Input validation
- Least-privilege database accounts
- Secure coding standards
- Application security testing

Evidence

Insert your real SQL injection screenshot.

Caption:

Figure 4: DVWA SQL injection test demonstrating unauthorized retrieval of multiple database records.

19. T2.5 — CROSS-SITE SCRIPTING

Objective

Determine whether user-controlled input is reflected into the browser without appropriate output encoding.

Payload

```
<script>alert('SecureCorp-XSS')</script>
```

Result

The JavaScript payload was executed in the browser during testing.

Security Impact

Reflected XSS may allow malicious JavaScript to execute in a victim's browser.

Potential consequences include:

- session theft
- malicious redirection
- phishing
- unauthorized browser actions
- manipulation of displayed content

Risk

High

Recommendation

Implement:

- context-aware output encoding
- input validation
- Content Security Policy
- secure cookie flags
- framework-level XSS protection

Evidence

Insert real XSS screenshot.

Caption:

Figure 5: Reflected XSS payload tested against the controlled DVWA application.

20. T2.6 — COMMAND INJECTION

Objective

Determine whether user input can cause unintended operating-system commands to execute.

Initial Input

127.0.0.1

Test Payload

127.0.0.1; whoami

Security Significance

Command injection is potentially one of the most serious application vulnerabilities because successful exploitation may allow an attacker to execute operating-system commands with the privileges of the vulnerable application.

Potential Impact

- Host compromise
- Data theft
- Malware installation
- Privilege escalation
- Lateral movement
- Destruction or modification of files

Risk

Critical

Recommendation

Avoid direct shell execution wherever possible.

Use:

- strict input allowlists
- safe APIs
- command argument separation
- application sandboxing
- least privilege
- secure coding reviews

Evidence

Insert the real command-injection screenshot.

21. T2.7 — FTP BRUTE FORCE

Objective

Perform controlled credential testing against the self-created low-privilege account.

Account

Username: securetest

Password: Secure123

Password List

123456

password

admin

Secure123

Tool

Hydra

Test

```
hydra -l securetest -P ftp-pass.txt ftp://192.168.137.72
```

A direct FTP login was also verified after correcting the account's home-directory configuration.

Security Impact

Weak FTP credentials can provide unauthorized access to files and potentially provide an initial foothold.

Recommendation

Prefer:

- SFTP
- strong authentication
- MFA where supported
- account lockout
- network restrictions
- disabling unnecessary FTP

Evidence

Insert:

- real FTP login screenshot
- real Hydra screenshot

Caption:

Figure 6: Controlled FTP authentication testing against the self-created laboratory account.

22. T2.8 — SSH BRUTE FORCE

Objective

Attempt controlled SSH credential testing against the self-created laboratory account.

Command

```
hydra -l securetest -P ftp-pass.txt ssh://192.168.137.72
```

Result

The legacy SSH service presented an incompatible MAC algorithm negotiation error:

kex error : no match for method mac algo

Therefore, the SSH brute-force exercise should be reported as:

Attempted – Not Successfully Completed

Do not write that the SSH password was successfully cracked.

Security Observation

The legacy SSH configuration itself represents a security and compatibility concern.

Recommendation

- Upgrade OpenSSH
 - Disable obsolete algorithms
 - Use modern cryptographic suites
 - Disable password authentication where practical
 - Use SSH keys
 - Restrict SSH by firewall
 - Implement MFA for administrative access
-

23. CONSOLIDATED ATTACK LOG

Use this table in the PDF.

ID	Activity	Target	Result	Detection
T2.1	Nmap	192.168.137.72	Completed	Not Visible
T2.2	DVWA mapping	DVWA	Completed	Manual
T2.3	DVWA brute force	DVWA	Tested successfully	Not Visible
T2.4	SQL Injection	DVWA	Demonstrated	Not Visible
T2.5	XSS	DVWA	Demonstrated	Not Visible
T2.6	Command Injection	DVWA	Tested	Not Visible
T2.7	FTP brute force	FTP	Controlled test completed	Not Visible
T2.8	SSH brute force	SSH	Attempted; MAC negotiation failure	Not Visible

Required Evidence Fields

For each attack, include:

- Attack ID
 - Date
 - Timestamp
 - Source IP
 - Target IP
 - Port
 - Service
 - Attack type
 - Tool
 - Payload/method
 - Result
 - Screenshot filename
 - Wazuh status
 - Analyst conclusion
-

24. WEEK 3 — DETECTION AND INVESTIGATION

Week 3 changes the viewpoint.

Instead of asking:

"Can I perform the attack?"

the analyst asks:

"Would a defender know that the attack occurred?"

The handbook specifically requires investigation of suspicious activity and distinction between Events, Alerts and Incidents. Home_lab_setup

25. EVENT, ALERT, INVESTIGATION AND INCIDENT

Event

An event is activity recorded by a system or log.

Example:

Failed SSH authentication

Alert

An alert occurs when a security monitoring system identifies suspicious activity.

Example:

Multiple failed authentication attempts detected

Investigation

An analyst examines:

- source IP
- destination IP

- timestamp
- username
- service
- frequency
- log source
- related activity

Incident

An incident is a confirmed security event that has sufficient impact or security significance to require response.

26. WAZUH DETECTION METHODOLOGY

The assessment uses three detection statuses.

Automatic

Wazuh generated an actual security alert supported by evidence.

Manual

The activity exists in logs but requires analyst investigation rather than an automatically generated alert.

Not Visible

Required target-side telemetry was not available, so detection cannot be confirmed.

For this project, **do not mark something Automatic merely because Wazuh was installed.**

27. DETECTION MATRIX

Activity	Expected Log Source	Detection Status	Explanation
Nmap	Network/host telemetry	Not Visible	Target telemetry unavailable

DVWA Brute Force	Apache/DVWA	Not Visible	Target web logs not ingested
SQL Injection	Apache/DVWA	Not Visible	Target application logs unavailable
XSS	Apache/DVWA	Not Visible	Target telemetry unavailable
Command Injection	Apache/DVWA/system	Not Visible	Target telemetry unavailable
FTP brute force	FTP/auth logs	Not Visible	Target logs unavailable
SSH brute force	auth logs	Not Visible	Target agent unavailable
Wazuh agent health	Wazuh	Automatic/Verified	WSL agent visibly Active

This is the technically defensible approach because the verified Wazuh agent was on WSL, not Metasploitable2. SecureCorp_COMPLETE_FINAL_PROJE...

28. ANALYST INVESTIGATION QUESTIONS

For every suspicious activity, the analyst should ask:

1. What happened?
 2. When did it happen?
 3. Which system was targeted?
 4. Which source generated the activity?
 5. Which destination was targeted?
 6. Which account was involved?
 7. Which service or application was targeted?
 8. Was the activity successful?
 9. Did Wazuh generate an alert?
 10. What containment or remediation action is required?
-

29. INCIDENT TIMELINE

Use a table like this:

Time	Activity	Evidence	Classification	Analyst Action
T1	Network discovery	Nmap output	Event	Review exposed services
T2	DVWA access	Browser screenshot	Event	Establish baseline
T3	Brute-force testing	Browser evidence	Suspicious Event	Review authentication controls
T4	SQL injection	DVWA result	Security Event	Assess database exposure
T5	XSS	Browser result	Security Event	Assess input/output validation
T6	Command injection	Application output	High-risk Event	Assess host execution risk
T7	FTP testing	FTP/Hydra evidence	Security Event	Review authentication
T8	SSH attempt	Hydra error	Event	Review legacy SSH configuration

Important conclusion

Because this was a controlled authorized lab exercise, these activities are not automatically "confirmed real-world incidents."

They are best classified as:

Controlled security events / assessment findings

unless actual compromise and impact are established.

30. FINDINGS REGISTER

Finding F-01 — Excessive Network Exposure

Severity: High

The target exposes a large number of services, including legacy protocols and remote-access services.

Impact: Increased attack surface.

Recommendation: Firewall unnecessary ports and disable unused services.

Finding F-02 — Legacy Telnet Exposure

Severity: Critical

Telnet provides insecure remote administration and can expose credentials.

Recommendation: Disable Telnet and replace it with secure SSH.

Finding F-03 — Exposed Database Service

Severity: High

MySQL was observed on port 3306.

Recommendation: Restrict database access to trusted application hosts.

Finding F-04 — SQL Injection

Severity: Critical

The DVWA application accepted a SQL injection payload and returned multiple records.

Recommendation: Parameterized queries and secure coding.

Finding F-05 — Cross-Site Scripting

Severity: High

The controlled application accepted and executed JavaScript input.

Recommendation: Output encoding, validation and CSP.

Finding F-06 — Command Injection

Severity: Critical

The vulnerable application allowed command-style input to be tested against the operating-system command interface.

Recommendation: Eliminate shell invocation and use safe APIs.

Finding F-07 — Weak Authentication Controls

Severity: High

Controlled brute-force testing demonstrated the importance of stronger authentication protections.

Recommendation: MFA, rate limiting, lockout and strong passwords.

Finding F-08 — Monitoring Coverage Gap

Severity: High

Wazuh was successfully deployed to WSL, but target-side monitoring of Metasploitable2 was not established.

Recommendation: Deploy an appropriate monitoring agent/log collection mechanism on the target and ingest authentication and web logs.

31. WEEK 4 — RECOMMENDATIONS

Recommendation 1 — Network Segmentation

Restrict access between users, servers, databases and administrative services.

Priority: Critical

Recommendation 2 — Remove Unnecessary Services

Disable:

- Telnet
- unused FTP
- VNC
- X11
- unnecessary RPC/NFS
- unused database listeners

Priority: Critical

Recommendation 3 — Patch Legacy Software

Replace obsolete versions of:

- OpenSSH
- Apache
- MySQL
- PostgreSQL
- BIND
- Samba
- FTP services

Priority: High

Recommendation 4 — Strong Authentication

Implement:

- strong passwords
- MFA
- rate limiting
- account lockout
- password rotation where appropriate

Priority: High

Recommendation 5 — Secure Web Development

Implement:

- parameterized SQL queries
- output encoding

- input validation
- secure APIs
- CSP
- secure session management

Priority: Critical

Recommendation 6 — Centralized Monitoring

Deploy Wazuh/log collection to relevant target systems.

Collect:

- `/var/log/auth.log`
- Apache access logs
- Apache error logs
- application logs
- FTP authentication logs

Priority: High

Recommendation 7 — Firewall

Restrict inbound ports to only services that are actually required.

Example production concept:

Internet
|
Firewall
|
+--- Web Server : 80/443
|
+--- SSH : restricted admin network
|
+--- Database : application subnet only

Recommendation 8 — Backups

Maintain:

- regular backups

- offline/isolated copies
 - tested restoration procedures
 - backup monitoring
-

Recommendation 9 — Security Awareness

Train users regarding:

- password security
 - phishing
 - MFA
 - suspicious activity reporting
 - safe handling of credentials
-

32. PRIORITIZED REMEDIATION ROADMAP

Timeline	Action	Priority
0–7 days	Restrict exposed services	Critical
0–7 days	Disable Telnet	Critical
0–7 days	Restrict MySQL	Critical
0–14 days	Fix SQL injection	Critical
0–14 days	Fix command injection	Critical
0–14 days	Fix XSS	High
0–30 days	Upgrade legacy services	High
0–30 days	Deploy target-side monitoring	High
0–30 days	Implement MFA	High
30–60 days	Improve segmentation	High
30–60 days	Security awareness	Medium

60–90
days

Continuous assessment

Medium

33. MANAGEMENT EXECUTIVE SUMMARY

What did we find?

The laboratory target presents a broad and outdated attack surface.

Why does it matter?

An attacker could potentially use exposed services, weak authentication or application vulnerabilities to gain unauthorized access.

Most important risks

1. SQL injection
2. Command injection
3. Legacy network services
4. Weak authentication
5. Exposed databases
6. Insufficient target-side monitoring

What should management do first?

1. Reduce exposed services.
 2. Patch obsolete software.
 3. Secure web applications.
 4. Strengthen authentication.
 5. Implement MFA.
 6. Improve monitoring.
 7. Segment critical systems.
-

34. FINAL ASSESSMENT CONCLUSION

The SecureCorp cybersecurity assessment successfully demonstrated the major phases of a practical security engagement.

The laboratory was built and assessed from an attacker's perspective using network discovery, application testing and controlled credential testing. The target exposed a significant number of legacy services, demonstrating the risks associated with excessive network exposure and outdated infrastructure.

DVWA testing demonstrated the security impact of common application vulnerabilities including authentication weaknesses, SQL injection, cross-site scripting and command-injection risks.

Wazuh Cloud was successfully established and the Ubuntu WSL agent was confirmed active and connected.

The principal limitation was monitoring coverage of the Metasploitable2 target. Since the verified Wazuh agent was installed on WSL rather than the target, target-side automatic detection should not be claimed without additional evidence.

The overall assessment therefore concludes:

The SecureCorp training environment demonstrates a high-risk legacy attack surface requiring significant hardening, while the monitoring environment provides a foundation for improved detection once target-side telemetry is fully deployed.

35. LIMITATIONS

Include this section. It actually makes your report more professional.

Limitation 1

Metasploitable2 is intentionally vulnerable and does not represent a modern production environment.

Limitation 2

Wazuh target-side monitoring was not fully established.

Limitation 3

Some legacy SSH cryptographic algorithms prevented completion of the SSH Hydra test.

Limitation 4

The assessment was performed in a controlled educational environment.

Limitation 5

The results should not be interpreted as evidence of compromise of a real SecureCorp production environment.

36. EVIDENCE INDEX

Use your **real screenshots first**.

Evidence	Description
E-01	Metasploitable2 login
E-02	Target IP/network
E-03	DVWA access
E-04	DVWA Low security
E-05	DVWA brute-force failed attempt
E-06	DVWA successful authentication
E-07	SQL injection
E-08	XSS
E-09	Command injection
E-10	FTP authentication
E-11	FTP Hydra
E-12	SSH Hydra attempt
E-13	Nmap discovery
E-14	Wazuh Cloud environment
E-15	Wazuh agent installation
E-16	Wazuh Active agent
E-17	Wazuh connected state

Then, **if you use the mock screenshots**, create a separate section:

SIMULATION / MOCK EVIDENCE

Never mix them into the real evidence section without a clear label.

37. APPENDIX — COMMANDS USED

Include the important commands.

Nmap

```
nmap -sV 192.168.137.72
```

Connectivity

```
nc -vz 192.168.137.72 22
```

FTP account

```
sudo useradd securetest  
sudo passwd securetest  
id securetest
```

FTP directory fix

```
sudo mkdir /home/securetest  
sudo chown securetest:securetest /home/securetest  
sudo chmod 755 /home/securetest
```

FTP password list

```
printf "123456\npassword\nadmin\nSecure123\n" > ftp-pass.txt
```

Hydra FTP

```
hydra -l securetest -P ftp-pass.txt ftp://192.168.137.72
```

Hydra SSH

```
hydra -l securetest -P ftp-pass.txt ssh://192.168.137.72
```

Wazuh status

```
sudo systemctl status wazuh-agent
```

Wazuh connection

```
sudo grep ^status /var/ossec/var/run/wazuh-agentd.state
```

PART B — 22-SLIDE PPT CONTENT

The PPT should **not contain the entire PDF text**. It should be concise and visual.

SLIDE 1 — TITLE

SECURECORP

Cybersecurity Assessment

Junior Cybersecurity Consultant – Four-Week Capstone

Prepared by: Navya.k

Add a professional cybersecurity background.

SLIDE 2 — EXECUTIVE SUMMARY

- Controlled cybersecurity assessment
 - Attacker + defender perspectives
 - WSL Ubuntu
 - Metasploitable2 + DVWA
 - Wazuh Cloud monitoring
 - Broad legacy attack surface identified
 - Multiple web security risks demonstrated
 - Major remediation opportunities identified
-

SLIDE 3 — PROJECT OBJECTIVES

- Build security laboratory
 - Discover target attack surface
 - Assess web application
 - Perform controlled attacks
 - Collect evidence
 - Monitor with Wazuh
 - Investigate suspicious activity
 - Assess risk
 - Recommend remediation
-

SLIDE 4 — METHODOLOGY

Put this large in the middle:

BUILD → DISCOVER → ATTACK → LOG → DETECT → INVESTIGATE → ASSESS → REMEDIATE → REPORT

Below it:

A complete cybersecurity operations lifecycle was followed throughout the engagement.

SLIDE 5 — LAB ARCHITECTURE

Show:

Ubuntu WSL
↓
Metasploitable2
↓
DVWA

Ubuntu WSL
↓
Wazuh Cloud

Add your architecture screenshot/diagram.

SLIDE 6 — ASSET INVENTORY

Table:

Asset	Role	Priority
Metasploitable2	Target	Critical
DVWA	Web target	Critical
WSL	Analyst	High
Wazuh	Monitoring	High
Network	Connectivity	High

SLIDE 7 — CIA ANALYSIS

Show:

Confidentiality — High

Integrity — High

Availability — High

Explain that target and monitoring assets require strong protection.

SLIDE 8 — NMAP DISCOVERY

Show the real Nmap screenshot.

Highlight:

21 FTP
22 SSH
23 Telnet
80 HTTP
139/445 SMB
2049 NFS
3306 MySQL
5432 PostgreSQL
5900 VNC
6000 X11

Key Finding

Excessive service exposure significantly increases attack surface.

SLIDE 9 — DVWA APPLICATION

Show real DVWA screenshot.

Modules:

- Brute Force
- SQL Injection
- XSS
- Command Injection

Security level:

LOW

SLIDE 10 — BRUTE FORCE

Show real screenshot.

Finding

Weak authentication can permit unauthorized login attempts.

Recommendations

- MFA
 - rate limiting
 - lockout
 - strong passwords
-

SLIDE 11 — SQL INJECTION

Show real SQLi screenshot.

Payload:

' OR '1'='1' #

Result

Multiple database records returned.

Severity

Critical

Fix

Parameterized SQL queries.

SLIDE 12 — XSS

Show real XSS screenshot.

Payload:

<script>alert('SecureCorp-XSS')</script>

Impact

JavaScript execution in victim browser.

Fix

Output encoding + CSP + validation.

SLIDE 13 — COMMAND INJECTION

Show real screenshot.

Payload:

127.0.0.1; whoami

Risk

Potential operating-system command execution.

Severity

Critical

SLIDE 14 — FTP / SSH TESTING

Split slide into two sections.

FTP

- Self-created `securetest`
- Controlled password list
- Hydra testing
- FTP login verified

SSH

- Hydra attempted
 - Legacy SSH MAC negotiation failed
 - No successful SSH brute-force claim
-

SLIDE 15 — WAZUH MONITORING

Show real Wazuh screenshots.

Results

- SecureCorp-Capstone created
 - SecureCorp-WSL enrolled
 - Wazuh 4.14.8
 - Agent Active
 - Agent connected
-

SLIDE 16 — DETECTION STATUS

Use:

Activity	Status
Nmap	Not Visible
DVWA Brute Force	Not Visible
SQLi	Not Visible
XSS	Not Visible
Command Injection	Not Visible
FTP	Not Visible
SSH	Not Visible
Wazuh Agent Health	Verified

Important

Target-side Wazuh telemetry was not established.

SLIDE 17 — INCIDENT INVESTIGATION

Show:

EVENT

↓

ALERT

↓

INVESTIGATION

↓

INCIDENT

Explain each term in one sentence.

SLIDE 18 — KEY FINDINGS

Critical

- SQL Injection
- Command Injection
- Telnet exposure

High

- Excessive network exposure
 - Weak authentication
 - Exposed MySQL
 - Legacy SSH
 - Monitoring gap
-

SLIDE 19 — RISK MATRIX

Create:

		IMPACT		
		Low	Med	High
Likelihood	High	M	H	C
	Medium	L	M	H
	Low	L	L	M

Place:

- SQLi → Critical

- Command Injection → Critical
 - Telnet → Critical
 - Exposed MySQL → High
 - Authentication → High
 - Monitoring gap → High
-

SLIDE 20 — RECOMMENDATIONS

1. Disable unnecessary services
 2. Restrict database access
 3. Upgrade legacy software
 4. Implement MFA
 5. Secure web applications
 6. Deploy target-side Wazuh
 7. Improve firewall rules
 8. Segment network
 9. Maintain backups
 10. Security awareness
-

SLIDE 21 — REMEDIATION ROADMAP

0–7 Days

- Disable Telnet
- Restrict exposed ports
- Restrict MySQL

0–30 Days

- Patch legacy systems
- Fix web vulnerabilities
- Improve authentication
- Deploy monitoring

30–90 Days

- Network segmentation
 - Continuous monitoring
 - Security awareness
 - Regular assessments
-

SLIDE 22 — CONCLUSION

Overall Assessment

High-risk legacy attack surface requiring immediate hardening.

Key Message

Reduce attack surface → strengthen authentication → secure applications →
improve monitoring → continuously assess.

Final Statement

The assessment demonstrated the complete cybersecurity consulting lifecycle from laboratory construction and attack-surface discovery through controlled security testing, monitoring, risk assessment and remediation planning.

Thank You

PART C — WHERE TO PUT YOUR REAL PHOTOS

For your **PDF**, I recommend approximately **35–45 pages** after inserting screenshots, tables and spacing.

Use this order:

PDF Section	Screenshot
Lab Setup	Metasploitable2
Network	Target IP
DVWA	DVWA home/login
Wazuh	Cloud environment
Wazuh	Agent installation
Wazuh	Agent Active
Wazuh	Connected state
Nmap	Nmap scan

Brute Force	Failed attempt
Brute Force	Successful attempt
SQLi	SQL injection result
XSS	XSS execution
Command Injection	Command result
FTP	FTP login
FTP	Hydra
SSH	Hydra error
Detection	Wazuh evidence
Remediation	Risk/recommendation table

The handbook's evidence guidance calls for screenshots that show enough context to identify the terminal/browser, target and relevant result, and its final checklist expects the network/application map, asset inventory, CIA, risk assessment, Wazuh evidence, attack log, detection notes, incident timeline, recommendations, executive summary and presentation.

Home_lab_setup